DATA VISUALIZATION FOR DECISION MAKING

PRACTICAL REPORT

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Course: M. SC.IT (Data Science)
Semester: 1
Course Code: P91A3DVD
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Practical 4: Data Transformation Using Power Query

Objective: Import data from multiple sources and clean, transform, and shape the data using Power Query Editor.

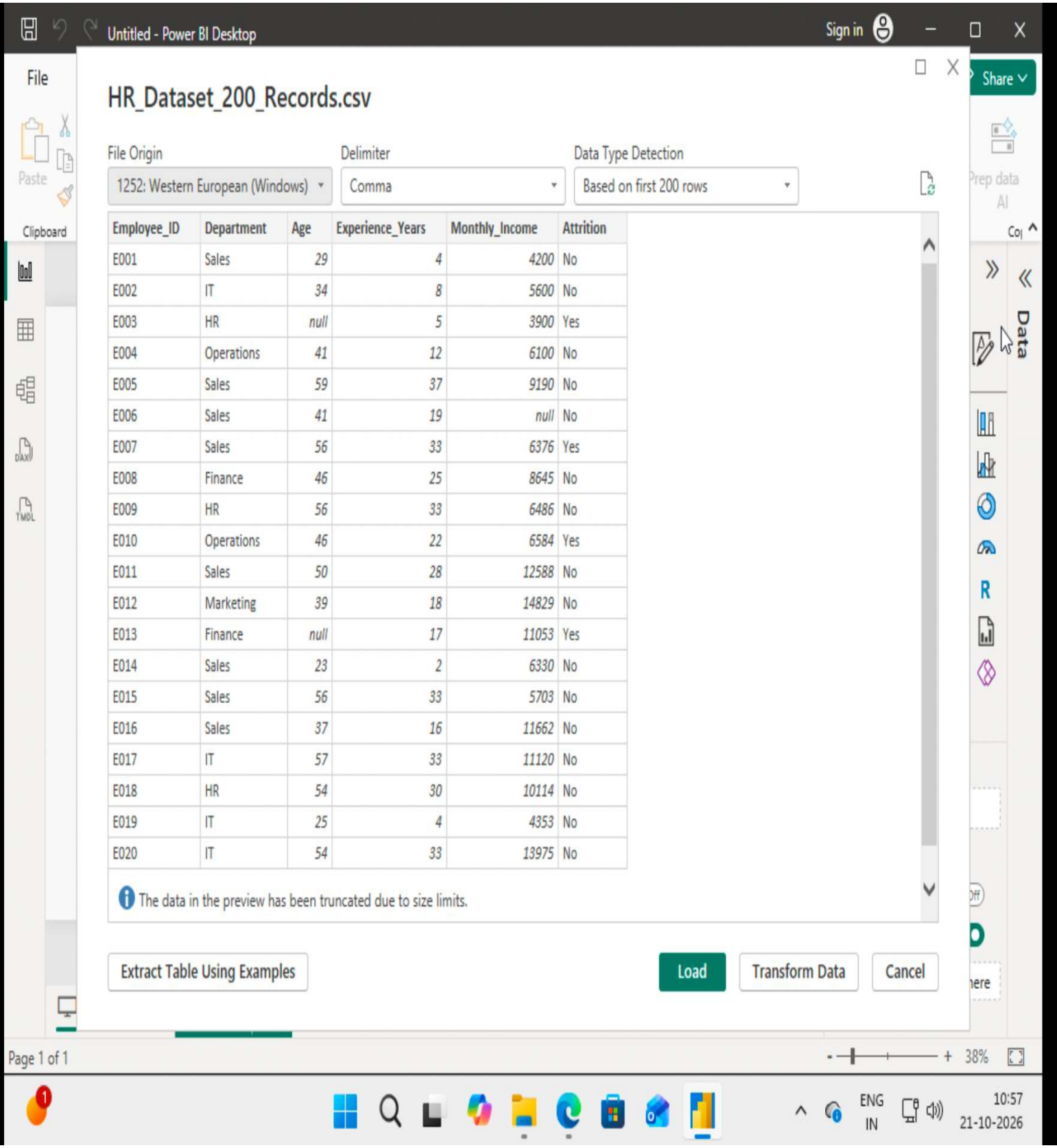
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Step 1: Get Data from Multiple Sources

• Open Power BI Desktop and select Home > Get Data.

• Choose Text/CSV to connect to HR_Dataset_200_Records.csv. Select the file and review its preview.

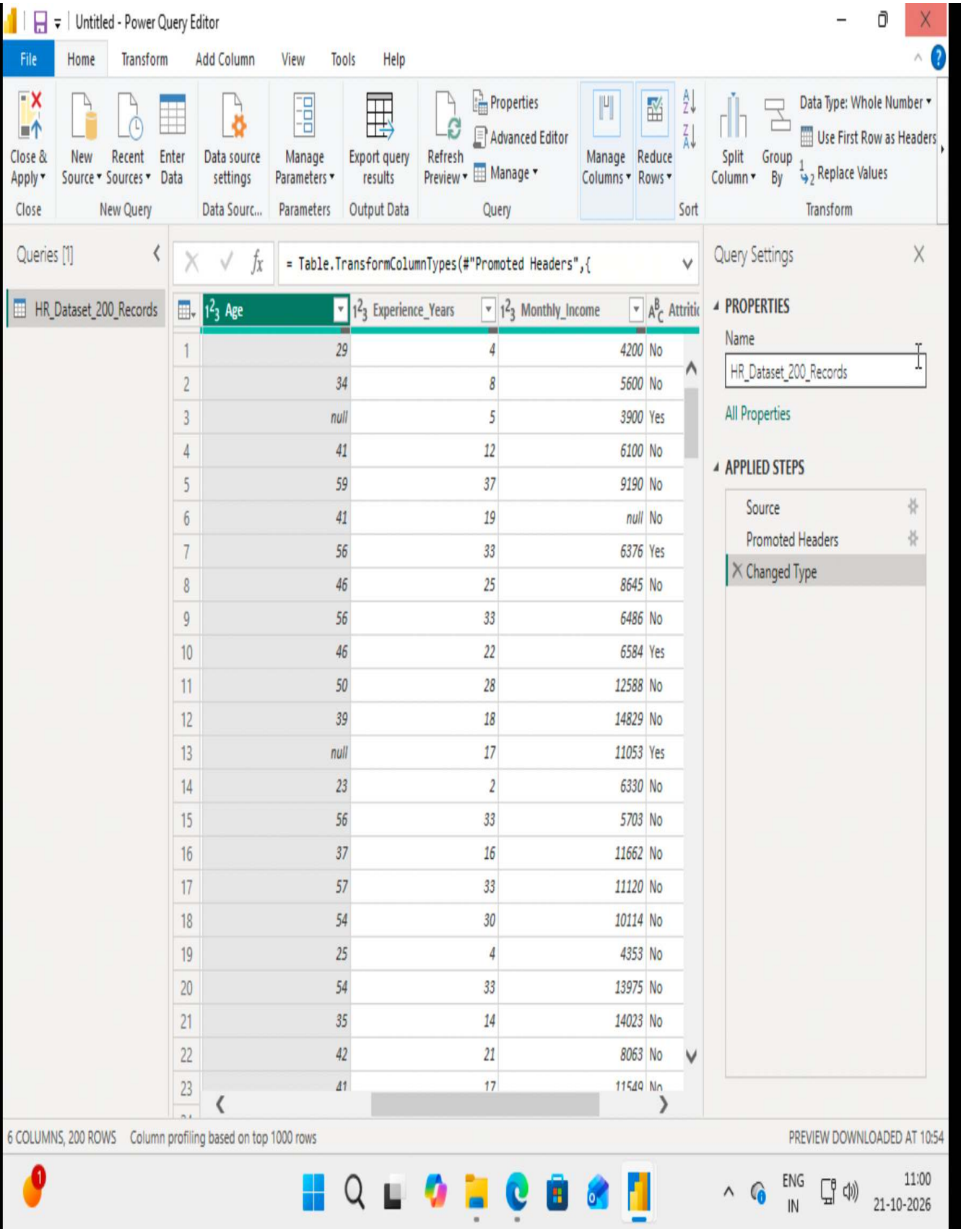
• Repeat Get Data and choose Excel Workbook if a second related dataset is available. Select the required sheet or table.



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Step 2: Open Power Query Editor

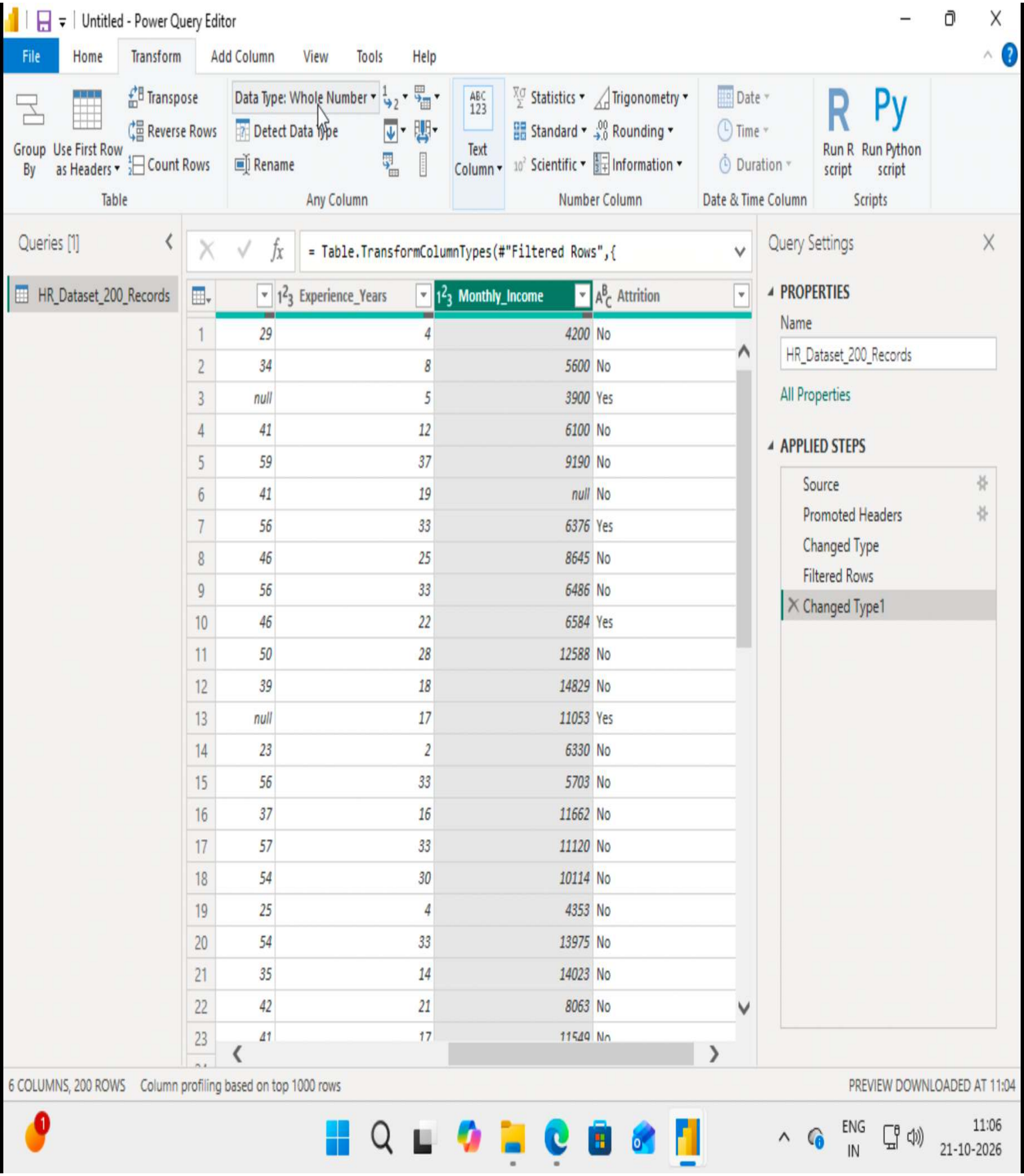
- In the preview / Navigator window, select Transform Data rather than Load.
- Power Query Editor opens with Queries on the left, data preview in the center, and Query Settings on the right.
- Select each query in turn to inspect column names and sample rows.



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Step 3: Check and Change Data Types

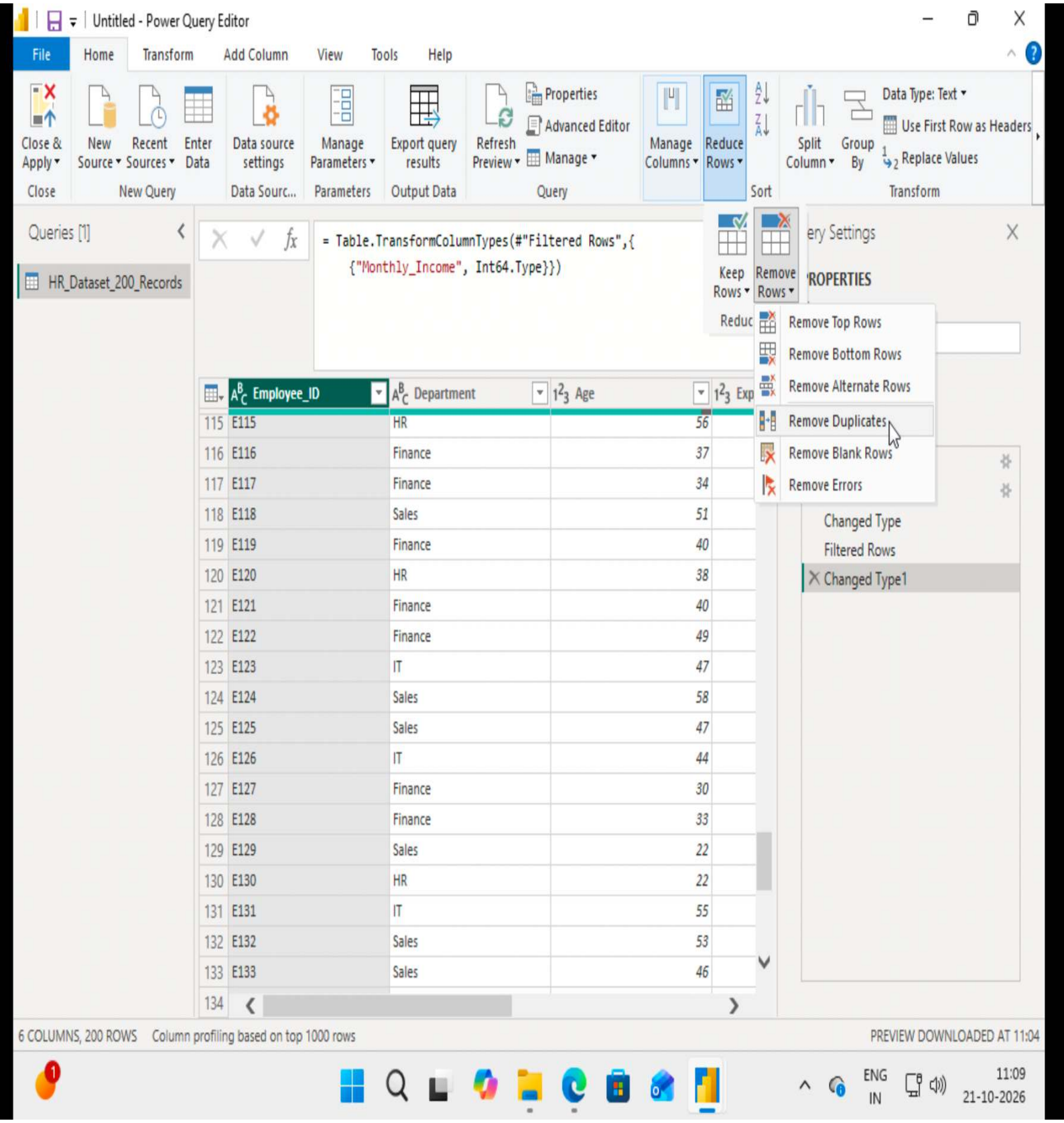
- Review the data type icon beside every column name.
- Set Employee_ID and Department to Text, Age and Experience_Years to Whole Number, and Monthly_Income to Whole Number or Decimal Number as appropriate.



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Step 4: Remove Duplicates and Handle Missing Values

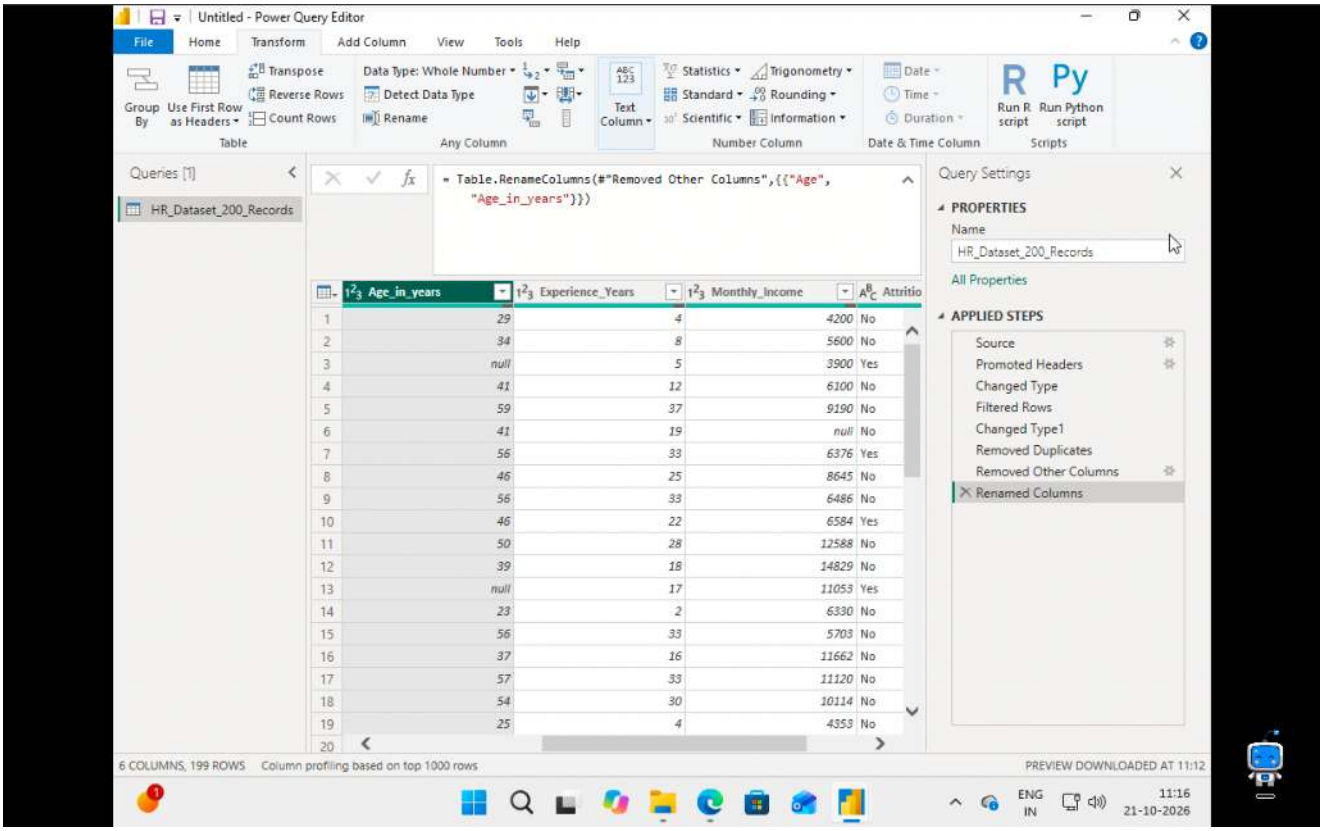
- Select a key column such as Employee_ID, or select the columns that define a duplicate record.
- Choose Home > Remove Rows > Remove Duplicates.
- For blanks or nulls, open the column filter and remove blank values when appropriate, or use Replace Values / Fill options if required.

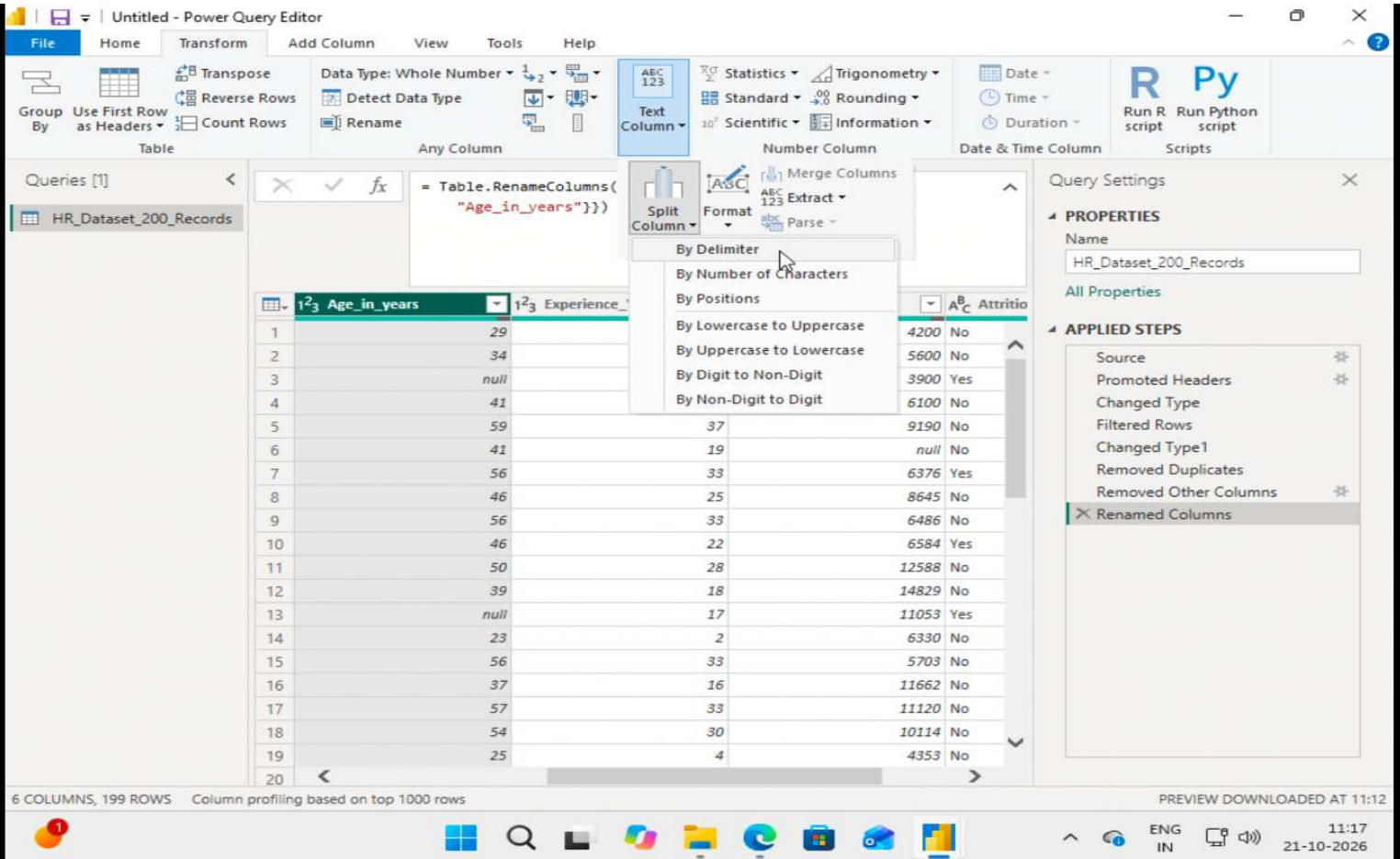


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Step 5: Rename Columns and Filter Rows

- Double-click a column heading or right-click it and choose Rename. Use clear, consistent names.
- Use the filter dropdown on Department, Attrition, or another relevant field to select categories.
- Sort a numeric column using Sort Ascending or Sort Descending. Each action appears under Applied Steps.

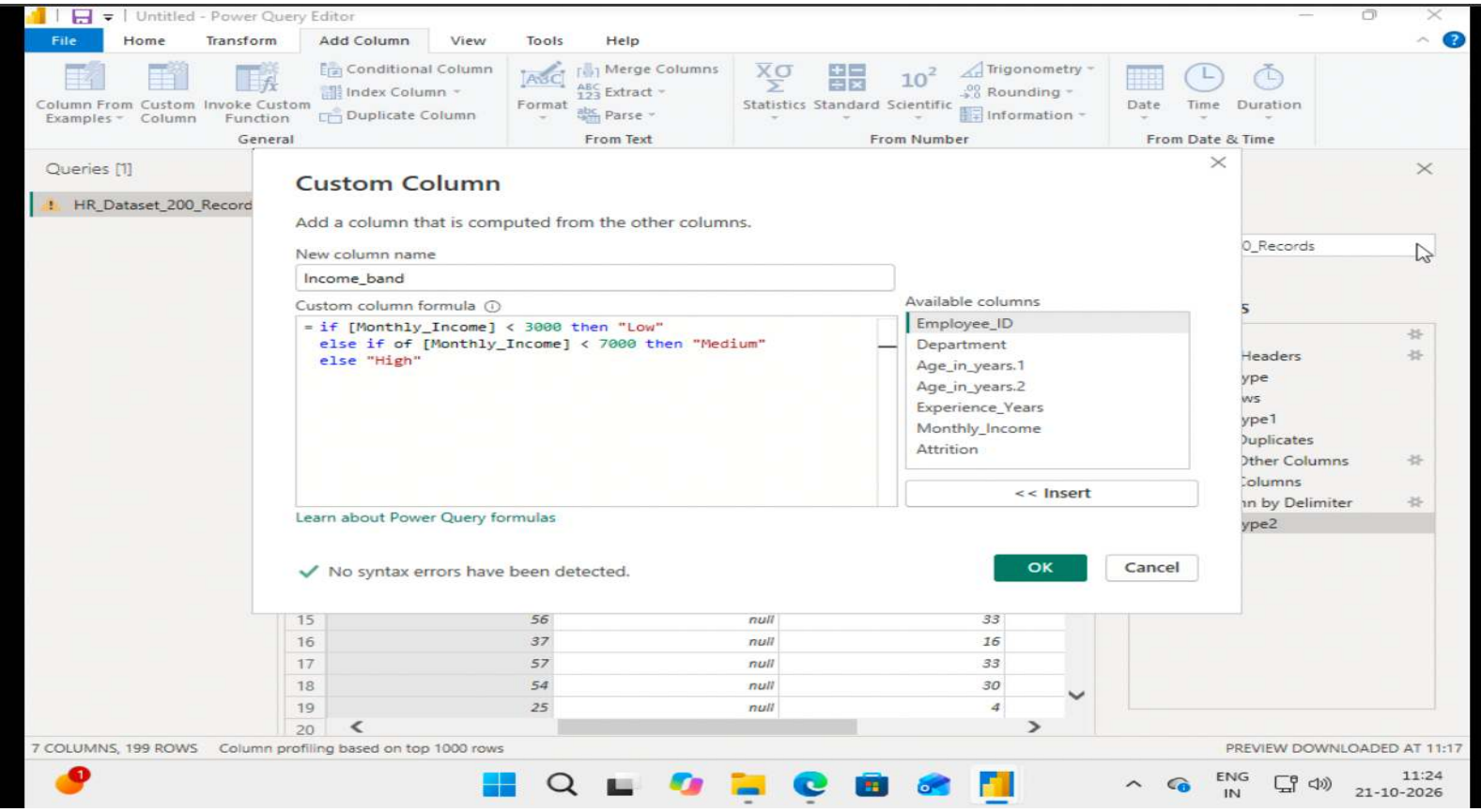


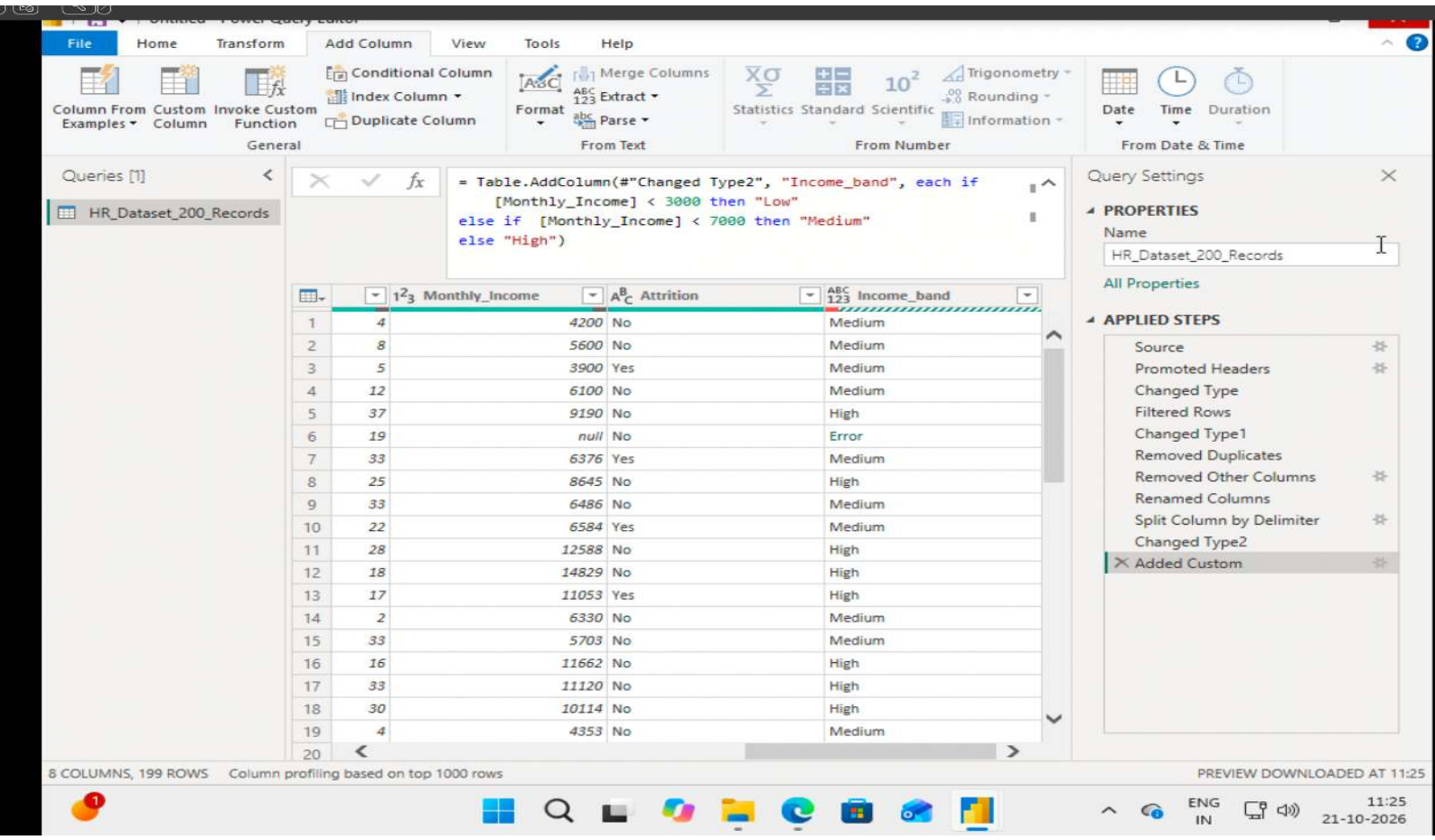


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Step 6: Split, Merge, or Add Columns

- To split a combined text field, select it and choose Transform > Split Column, then choose a delimiter.
- To create a derived field, choose Add Column > Custom Column and enter a suitable formula.
- For example, an Income Band column can categorize income values if this is useful for your dataset.



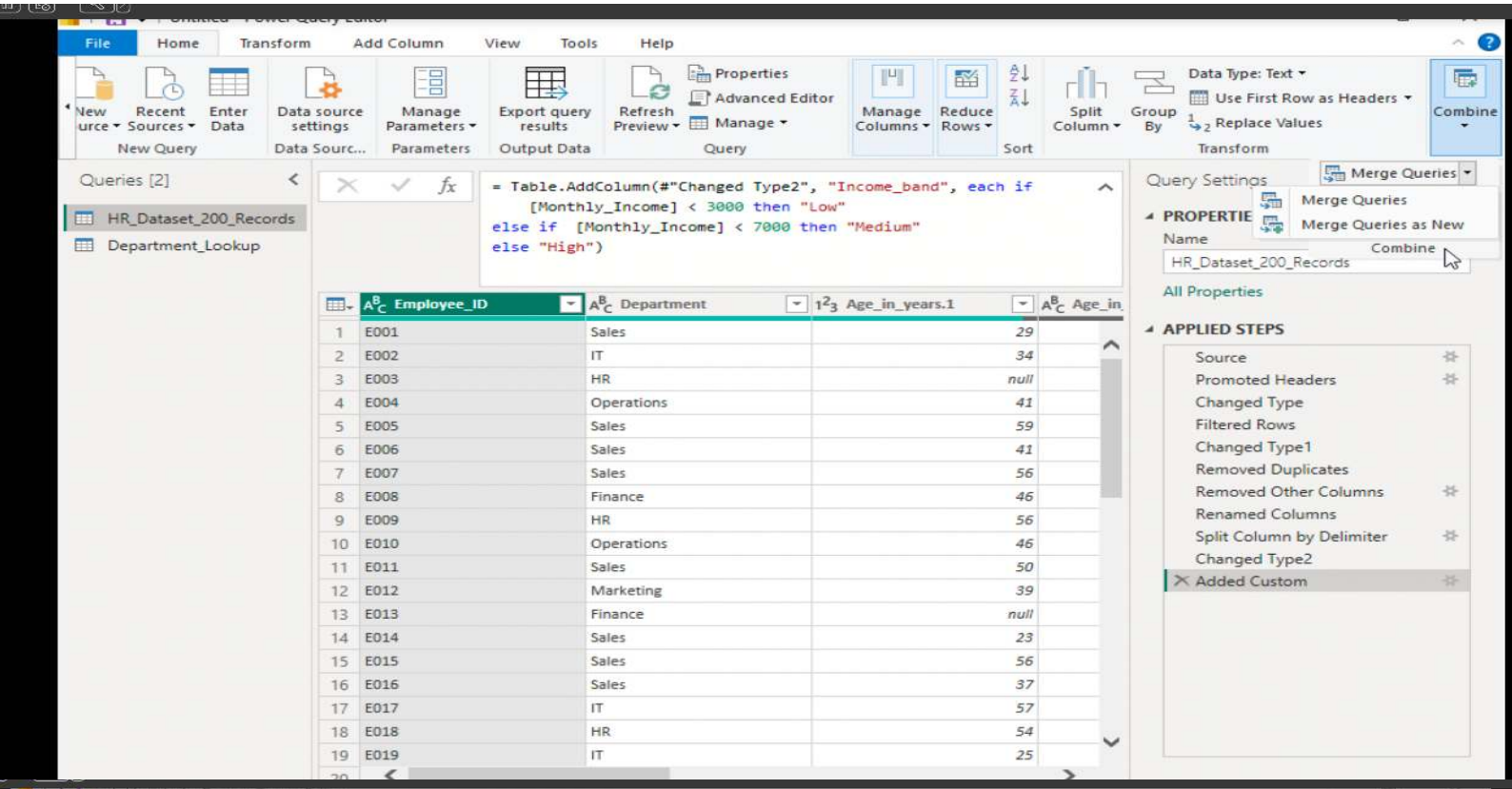


	Monthly_Income	Attrition	Income_band
1	4200	No	Medium
2	5600	No	Medium
3	3900	Yes	Medium
4	6100	No	Medium
5	9190	No	High
6	null	No	Error
7	6376	Yes	Medium
8	8645	No	High
9	6486	No	Medium
10	6584	Yes	Medium
11	12588	No	High
12	14829	No	High
13	11053	Yes	High
14	6330	No	Medium
15	5703	No	Medium
16	11662	No	High
17	11120	No	High
18	10114	No	High
19	4353	No	Medium

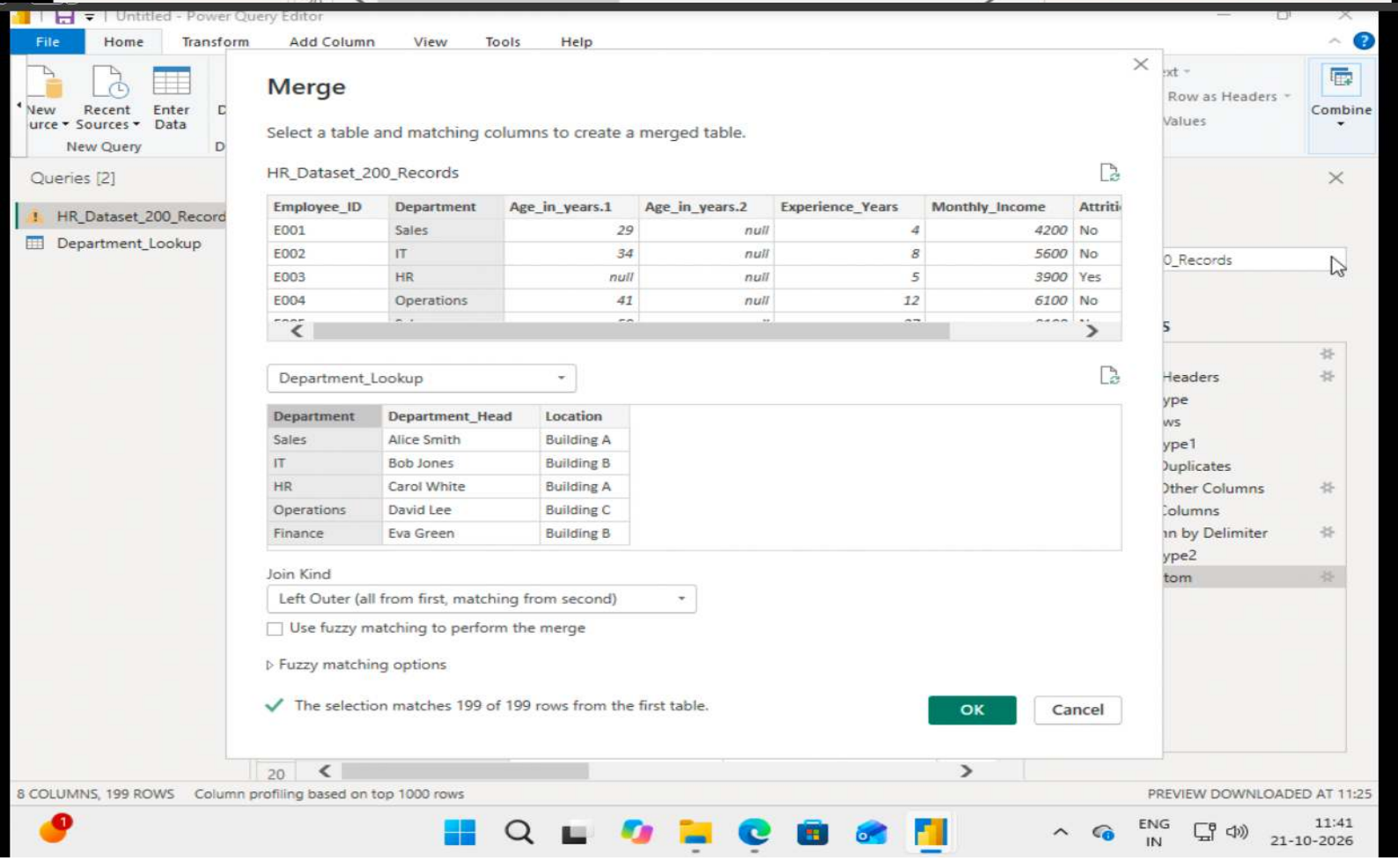
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Step 7: Combine Queries (Optional)

- If the sources share a key, select Home > Merge Queries > Merge Queries as New.
- Choose the primary table and lookup table, then select the matching key column in both tables.
- Use Left Outer join to keep every row from the first table and bring in matching rows from the second. Select OK, then expand the new column.



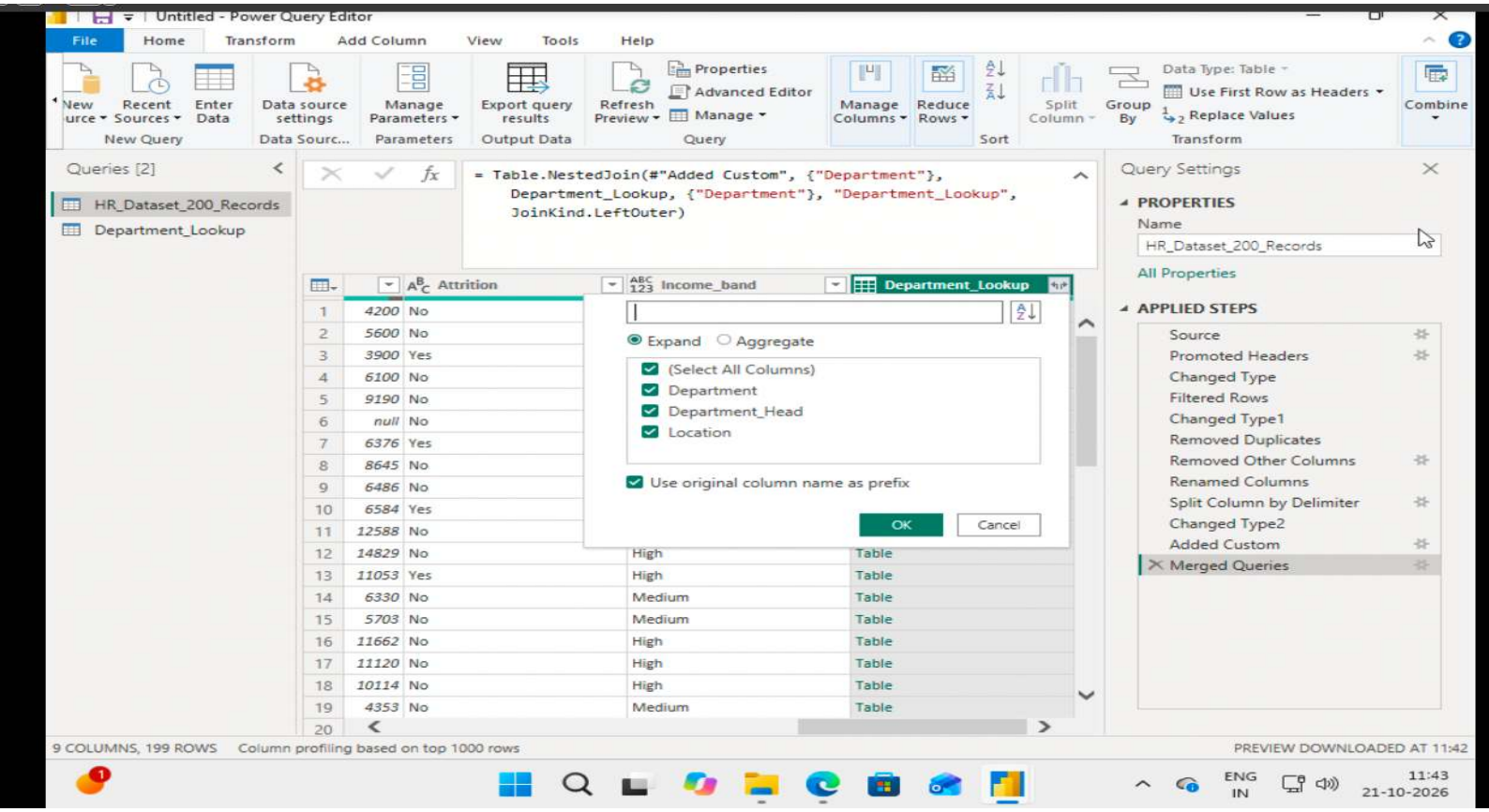
	Employee_ID	Department	Age_in_years.1	Age_in
1	E001	Sales	29	
2	E002	IT	34	
3	E003	HR	null	
4	E004	Operations	41	
5	E005	Sales	59	
6	E006	Sales	41	
7	E007	Sales	56	
8	E008	Finance	46	
9	E009	HR	56	
10	E010	Operations	46	
11	E011	Sales	50	
12	E012	Marketing	39	
13	E013	Finance	null	
14	E014	Sales	23	
15	E015	Sales	56	
16	E016	Sales	37	
17	E017	IT	57	
18	E018	HR	54	
19	E019	IT	25	

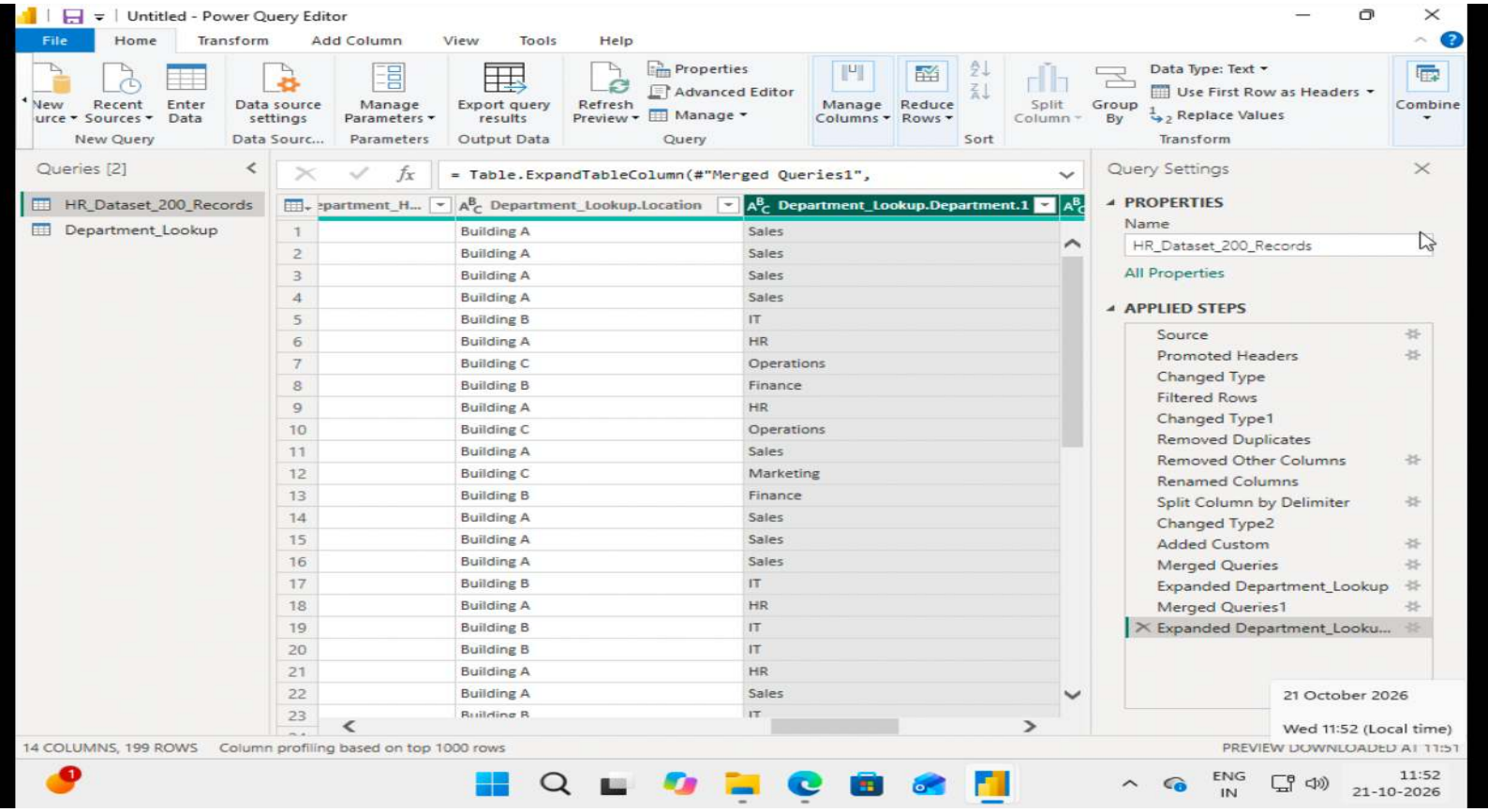


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Step 8: Close & Apply

- Review the final preview and Applied Steps list for each query.
- Select Home > Close & Apply.
- Power BI applies the transformations and loads the resulting query tables into the data model.

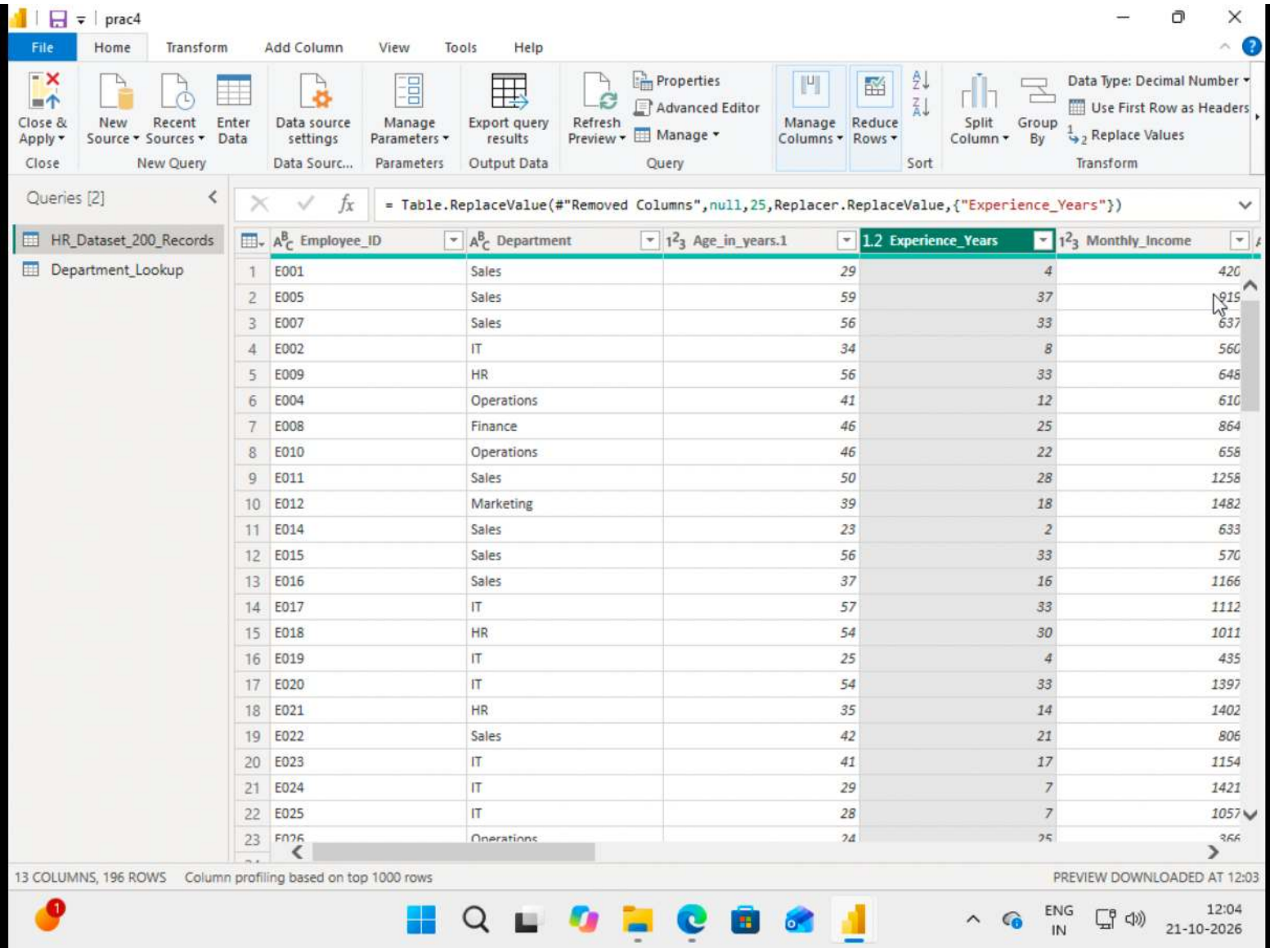




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Step 9: Verify the Transformed Data

- Switch to Data view in Power BI Desktop and select the loaded table.
- Confirm that expected columns are present, data types are correct, duplicates have been handled, and filters or new columns behave as intended.
- Save the Power BI file using File > Save As.



Result

Data from the selected source files was connected in Power BI Desktop and prepared using Power Query Editor. The practical demonstrated data type correction, duplicate and missing-value handling, column renaming, filtering, sorting, optional column shaping, and optional query combining. The transformed data was applied and verified in Power BI Desktop.

